

Multiple Choice (10 marks)

1. Which measurement would best be rounded to the nearest thousand?
 - (a) The number of yards run by a player in a game
 - (b) The number of people at a professional football game
 - (c) The length of a football in inches
 - (d) The weight of a professional football player in pounds
 - (e) The number of touchdowns in a professional football season
2. Maddie will ride her bike a total of 56 miles over 7 days. She will ride the same number of miles each day. What is the total number of miles Maddie will ride each day?
 - (a) 9
 - (b) 63
 - (c) 56
 - (d) 8
 - (e) 28
3. Let $y = f(x)$ be a solution of the differential equation $x \, dy + (y - xe^x) \, dx = 0$ such that $y = 0$ when $x = 1$. What is the value of $f(2)$?
 - (a) $2/e$
 - (b) $e^2 + 2$
 - (c) $1/(2e)$
 - (d) $e^2/2$
 - (e) $2/(e^2)$
4. Consider the two sets $X = \{10, 30, 45, 50, 55, 70, 90\}$ and $Y = \{10, 30, 35, 50, 65, 70, 90\}$. Which of the following is false?
 - (a) The sets have identical medians.
 - (b) The sets have identical means.
 - (c) The sets have identical ranges.
 - (d) None of the above are false.
5. Tara can develop 2 rolls of film in about 18 minutes. At this rate, how long will it take her to develop 8 rolls of film?
 - (a) 1 hour 44 minutes

- (b) 1 hour 20 minutes
- (c) 1 hour
- (d) 1 hour 12 minutes
- (e) 1 hour 30 minutes

True / False (10 marks)

Answer True or False

6. True or False: The number 47,520, when rounded to the nearest thousand, results in 47,000.
7. True or False: If A and B are orthogonal square matrices with $\det A = -\det B$, then $\det(A+B)$ must be equal to -2.
8. True or False: The greatest common factor of 36 and 90 is 30.
9. True or False: In kite WXYZ, the length of side ZY is 25.3 units, derived from the properties of the diagonals and segment lengths.
10. True or False: The positive square root of 200 is approximately 11% larger than the positive square root of 121.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given the parametric equations $x = 4 \cos(t)$ and $y = 8 \sin(x)$, determine the second derivative y''_{xx} at $t = \frac{\pi}{3}$ and discuss its significance in the context of curvature.
12. Using Euler's method, analyze the steps to approximate the solution for the differential equation $dy/dx = y^2 e^x$ at $x = 6$ with the initial condition $y(0) = 0.01$ and step size $h = 0.5$. What is the resulting value of $y(6)$, and discuss the implications of your findings?

End of Paper